RNCP LAB | Production Deployment Simulation Extra

Objectives

- Set up a blue-green deployment using AWS CodeDeploy.
- Perform a rollback in case of issues.

AWS CodeDeploy

AWS CodeDeploy automates the deployment of applications to various compute services such as Amazon EC2 instances. It ensures reliable and consistent deployments, reducing the risk of errors.

Rollback

A rollback reverts your application to a previous stable version if a new deployment encounters issues, maintaining application stability and user experience.

Blue-Green Deployment

Blue-Green Deployment involves two identical environments:

- **Blue:** Current live environment.
- **Green:** New environment where the updated application is deployed.

Traffic is switched from Blue to Green after successful deployment and testing, minimizing downtime and allowing easy rollback if needed.

Tasks

1. **Set Up Blue-Green Deployment**
2. **Deploy Application**
3. **Perform Rollback (if necessary)**

Prerequisites

- **AWS Account:** Ensure you have access with permissions for CodeDeploy, EC2, and IA

Ask AI Tutor

- **Source Code Repository:** A repository (e.g., GitHub) with your application code.
- **Application Code:** Includes an `appspec.yml` file and any necessary deployment scripts.

1. Set Up Blue-Green Deployment

Step 1: Navigate to AWS CodeDeploy

1. Log in to AWS Management Console.
2. Access **CodeDeploy**:
   - Click on "Services" in the top-left corner.
   - Under "Compute", select "CodeDeploy".

Step 2: Create a New Application

1. **Create Application**:
   - In the CodeDeploy dashboard, click "Create application".
   - Application name: `BlueGreenApp`
   - Compute platform: Select **"EC2/On-premises"**.
   - Click **"Create application"**.

Step 3: Create Deployment Group

1. **Initiate Deployment Group Creation**:
   - Within `BlueGreenApp`, click "Create deployment group".
2. **Configure Deployment Group**:
   - Deployment group name: `BlueGreenDeployment`
   - Service role: Select an existing IAM role with CodeDeploy permissions or create a new one.
   - Deployment type: Choose **"Blue/Green"**.
3. **Environment Configuration**:
   - **Blue environment:** Select your current EC2 instances (live environment).
   - **Green environment:** Select EC2 instances for the new deployment (idle environment).
   - Load balancer: If using one, select the appropriate load balancer to manage traffic.
4. **Create Deployment Group**:
   - Click **"Create deployment group"**.

2. Deploy Application

Step 1: Prepare Application Revision

1. **Ensure `**appspec.yml**` Exists**:

- Example:

```
version: 0.0
os: linux
files:
- source: /
destination: /var/www/html
hooks:
AfterInstall:
- location: scripts/restart_server.sh
timeout: 300
runas: root
```

2. **Create Deployment Script**:

- Example `restart_server.sh`:

```
#!/bin/bash
sudo systemctl restart apache2
```

3. **Make Script Executable**:

```
chmod +x scripts/restart_server.sh
```

Step 2: Push to Repository

1. **Commit and Push Changes**:

```
git add .
git commit -m "Add appspec.yml and deployment scripts"
git push origin main
```

2. **Ensure CodeDeploy Access**:
   - If using GitHub, authorize CodeDeploy to access your repository.
   - Alternatively, upload the application revision to an S3 bucket.

Step 3: Create a Deployment

1. **Initiate Deployment**:
   - In the CodeDeploy console, go to `BlueGreenApp` and select `BlueGreenDeployment`.
   - Click **"Create deployment"**.
2. **Configure Deployment Details**:
   - Revision type: Select your repository type (e.g., GitHub or S3).
   - Revision location: Provide the repository details or S3 bucket information.
   - Deployment type: Ensure **"Blue/Green"** is selected.
3. **Start Deployment**:
   - Click **"Create deployment"** to begin.
4. **Monitor Deployment**:
   - Wait for the deployment status to change to **"Succeeded"**.
5. **Verify Deployment**:
   - Access your application URL to ensure the new version is live.

3. Perform Rollback (if necessary)

Step 1: Identify Issues

- If after deployment you encounter issues (e.g., application errors), proceed to rollback.

Step 2: Initiate Rollback

1. **Access CodeDeploy Console**:
   - Navigate to CodeDeploy > `BlueGreenApp` > `BlueGreenDeployment`.
2. **Locate Current Deployment**:
   - Go to the **"Deployments"** tab.
   - Select the problematic deployment.
3. **Stop Deployment**:

- Click **"Actions" > "Stop deployment"**.

4. **Choose Rollback Option**:
   - Select **"Rollback to previous version"** (Blue environment).
   - Confirm the rollback.
5. **Monitor Rollback**:
   - Ensure the rollback status updates to **"Succeeded"**.
6. **Verify Rollback**:
   - Access your application to confirm it's reverted to the previous stable version.

Submission

Provide Configuration Files

1. **`**appspec.yml`** :

Copy Explain

```
version: 0.0
os: linux
files:
- source: /
destination: /var/www/html
hooks:
AfterInstall:
- location: scripts/restart_server.sh
timeout: 300
runas: root
```

2. **Deployment Script (`**restart_server.sh**`)**:

```
#!/bin/bash

sudo systemctl restart apache2
```

Copy Explain

Document the Rollback Process

- **Screenshots**:
  - Capture steps taken in CodeDeploy to initiate and complete the rollback.
  - Include images showing the deployment status before and after rollback.
- **Logs**:
  - Provide any relevant logs indicating a successful rollback.

Compile Your Work

1. **Create a Document**:
   - Use a Word or PDF document to include all configuration files, screenshots, and descriptions.
2. **Organize Sections**:
   - Introduction
   - Configuration Files
   - Deployment Steps
   - Rollback Steps
   - Screenshots and Logs
3. **Upload and Share**:
   - Upload the document to a cloud storage service (e.g., Google Drive, Dropbox).
   - Ensure the link is set to **"Anyone with the link can view"**.
   - Submit the shareable link as directed.

Summary

This simplified lab guides you through setting up a blue-green deployment using AWS CodeDeploy and performing a rollback if needed. By following these steps, you can manage application deployments efficiently, ensuring minimal downtime and maintaining application stability.

RNCP LAB | Production Deployment Simulation

Extra

URL

Enter the URL here

Submit lab